DATA SECURITY

Unblock Revenue from Security Bottlenecks and Protect Cash Flow from Compliance Fines

Whether you're blocked by security reviews, losing engineering capacity to duplicated protection and data quality issues, or exposed to regulatory fines, success requires centralized control: what's encrypted, what's validated, and who can access it.

Your data is in Kafka. But is it protected?

We see the same pattern across industries: sensitive data flows through Kafka in plaintext, application teams build their own encryption and validation with implementations that vary by language, and non-compliant data slips into production undetected. The result is projects that stall, budgets that balloon, and compliance gaps that widen with every new team. Conduktor enforces encryption, data quality rules, and access controls at the wire level, so compliance is provable, not aspirational.

01

Revenue is stuck waiting on security approvals

Every data initiative touching sensitive fields sits in security review for months, delaying revenue-generating features and partnerships.

No centralized way to prove encryption is enforced, so InfoSec requires manual verification for every new application and data stream.

Engineers sit idle waiting on approvals while legal and security teams work through a backlog they can never clear fast enough.

With Conduktor
Centralized encryption means InfoSec signs off once, not per team. Integrates with your existing KMS so there's nothing new to approve.

02

Margins shrink from duplicated work and bad data reaching production

Custom encryption implementations across dozens of teams cost $200K+ annually in duplicated engineering effort and ongoing maintenance.

Each team implements encryption and validation differently, with no centralized standard, so non-compliant data slips through and breaks downstream systems.

Developers spend weeks building protection that already exists elsewhere, and one missed field or malformed message creates an organization-wide incident.

With Conduktor
Encryption and data quality rules enforced once at the gateway across all clusters and languages. Zero code changes, no per-team implementation.

03

Cash flow is one data breach away from fines

Non-compliance with GDPR can cost up to 4% of global revenue, HIPAA violations up to $2M per category, and PCI-DSS fines that compound monthly.

Kafka ACLs operate at the topic level, with no native ability to mask or restrict specific fields, so sensitive and non-sensitive data are equally exposed.

Security teams carry the liability but have no tools to restrict access, so every incident becomes their failure to contain.

With Conduktor
Field-level encryption, dynamic masking, and tokenization ensure sensitive data is never exposed, even if access controls are breached.

IMPACT WITH CONDUKTOR

UP TO
85%
Faster Security Implementation
Per project vs. custom build

OVER
$200K
Annual Savings
From efficient engineering and prevented downtime

AVERAGE OF
500hrs
Saved Per Year
Across implementation, maintenance, and audit prep

REDUCE
Days
to Production
For new data streams with sensitive data

HEALTHCARE & INSURANCE

Major US Healthcare Organization

Centralized Encryption Across 150+ Applications Processing Protected Health Information

A major US healthcare organization had 150+ Kafka applications streaming protected health information with no centralized encryption. Each team encrypted differently across Java, Python, and .NET. When a team missed a field, there was no fallback. InfoSec required proof of end-to-end encryption before approving new data streams, creating 4-6 month delays on every project.

With Conduktor Gateway, they enforced field-level encryption across all 150+ applications with zero code changes. Policies defined once, enforced centrally, with Vault integration for key management. New data streams now go to production in under three weeks.

150+	4mo → 3wks	$250K+	Zero
Apps protected	Time to production	Annual savings	Code changes required

"Every new data stream required a 4-month security review because we couldn't prove encryption was enforced. Now it's a configuration, not a project."

Director of Platform Engineering, Major US Healthcare Organization

How We Do It

Conduktor Gateway sits between your applications and Kafka, enforcing encryption, masking, and access controls at the wire level without touching application code.

Data Confidentiality

- **Centralized Encryption**
  Unified KMS across clusters
- **Multi-cloud KMS Integration**
  Works across AWS, Azure, GCP, and Vault
- **Field-level Encryption**
  Protect sensitive fields and keep data usable
- **Tokenization**
  Mask sensitive data without breaking analytics

Platform Guardrails

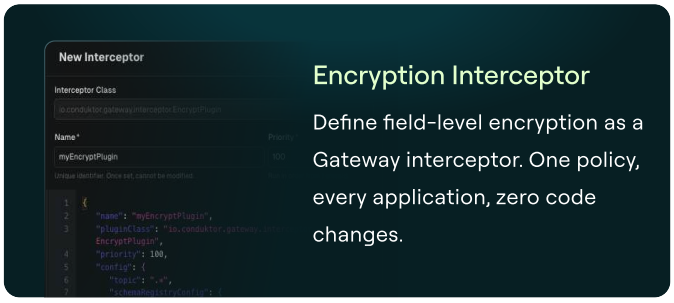
- **Configuration Guardrails**
  Enforce naming, retention, and schema rules
- **Traffic Control & Throttling**
  Rate limits and throttling per tenant or cluster
- **Producer & Consumer Policies**
  Control reliability and access
- **Header & Message Handling**
  Modify headers without application changes

Data Quality

- **Detection & Enforcement**
  Evaluate data against validation rules and block or mark non-compliant messages before they reach Kafka.

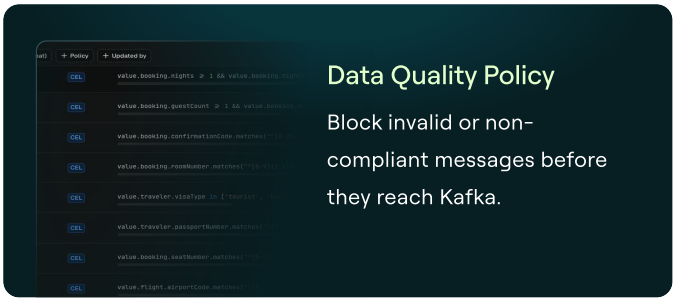
Data Integrity

- **Cryptographic Signing**
  Sign messages at the gateway to ensure integrity and verify message origin for compliance.



Encryption Interceptor

Define field-level encryption as a Gateway interceptor. One policy, every application, zero code changes.



Data Quality Policy

Block invalid or non-compliant messages before they reach Kafka.

Ready to protect your Kafka data?

Watch a 5 minute demo to see how Conduktor secures sensitive data streams without code changes.

See it in action now